Math 252 [redacted] **Quiz 3** **Name:**

1. Use the **limit of Riemann sum definition** of the definite integral to evaluate $\int_0^2 (4 + 3x^2)\,dx$.

Note: You may use right-endpoint Riemann sum. No credit will be given if you use other method.

Basic Summation Formulas: $\sum_{i=1}^{n} c = cn$, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, $\sum_{i=1}^{n} i^2 = \frac{2n^3 + 3n^2 + n}{6}$

$$\sum_{i=1}^{n} i^3 = \frac{n^4 + 2n^3 + n^2}{4}$$

2. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of $F(x) = \int_0^{\ln x} \sqrt{t^4 + 1}\,dt$.